Lab Session - Week 9

Mina Malvandy

2024-04-18

R Markdown

First you should save all the data in your own working directory. In order to find your current working directory, run the following:

```
getwd()
```

```
## [1] "/cloud/project"
```

This is my working directory ("/cloud/project") which is probably different from yours.

This week, we will learn how to perform hypothesis tests about the relation between two variables.
We will try three tests for different measurement level combinations:
1. nominal/ordinal - nominal/ordinal 2. nominal/ordinal - continuous
3. continuous - continuous

Note: We will assume throughout that the data comes from an SRS, but this will not be true.
We do this to simplify the calculations. We will take into account weights at certain points however.

1. LOAD THE DATA

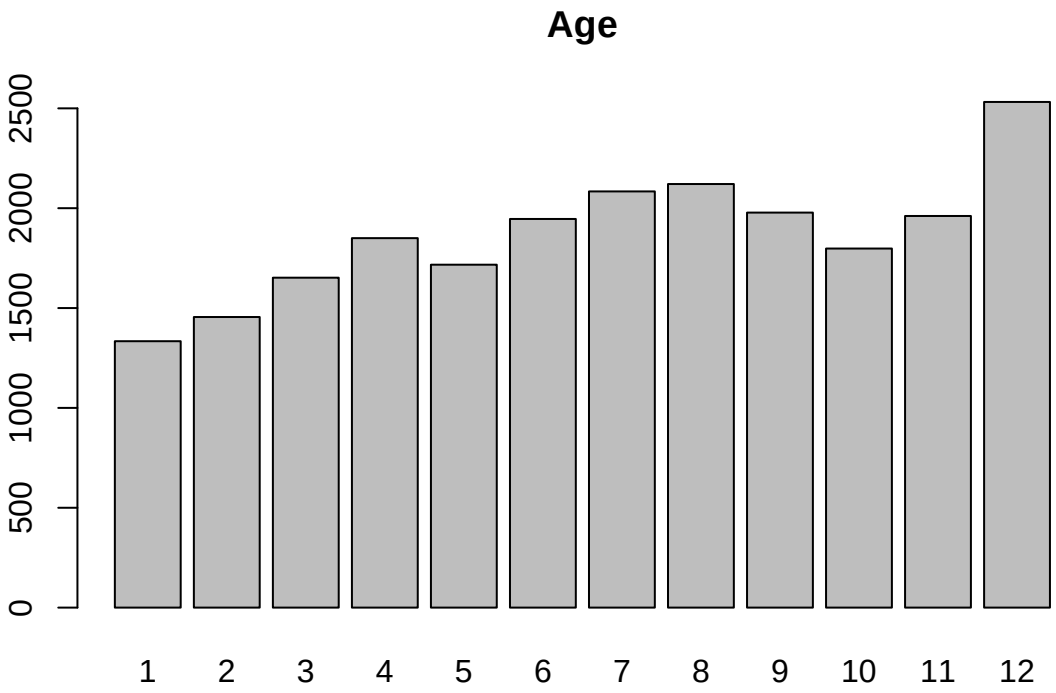
```
load("/cloud/project/LFSdata.RData")
```

```
table(data$AGEEULR)
```

Let's look at Age.

```
##
##    1    2    3    4    5    6    7    8    9   10   11   12
## 1334 1455 1652 1850 1717 1946 2084 2121 1978 1798 1961 2532
```

```r
barplot(table(data$AGEEULR), main="Age") # the 12 age bands
```

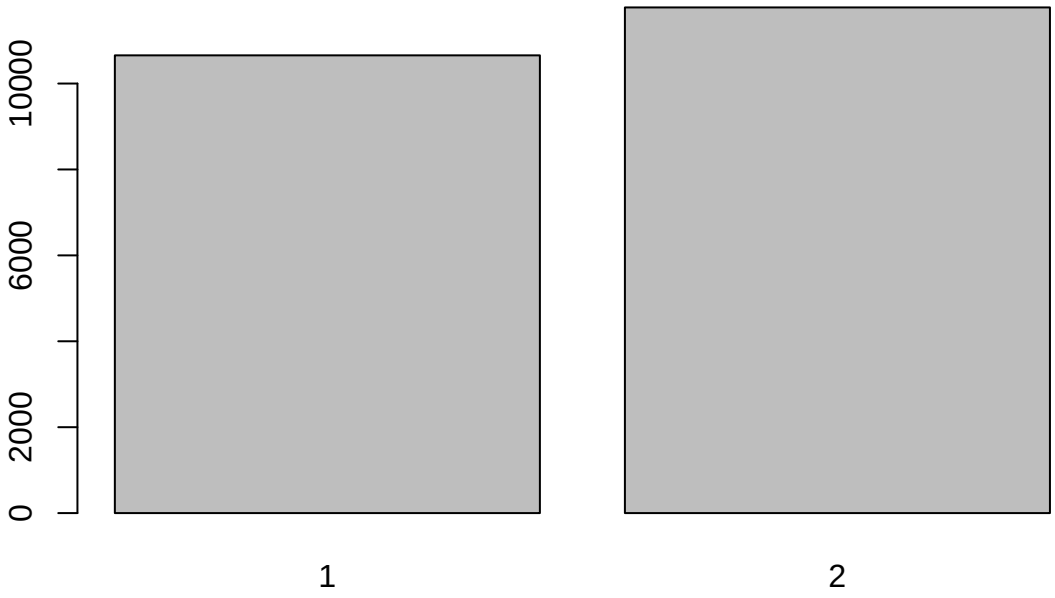


Let's look at Sex.

```r
table(data$SEX)
```

```
##
##     1     2
## 10656 11772
```

```r
barplot(table(data$SEX), main="SEX") # two sexes
```

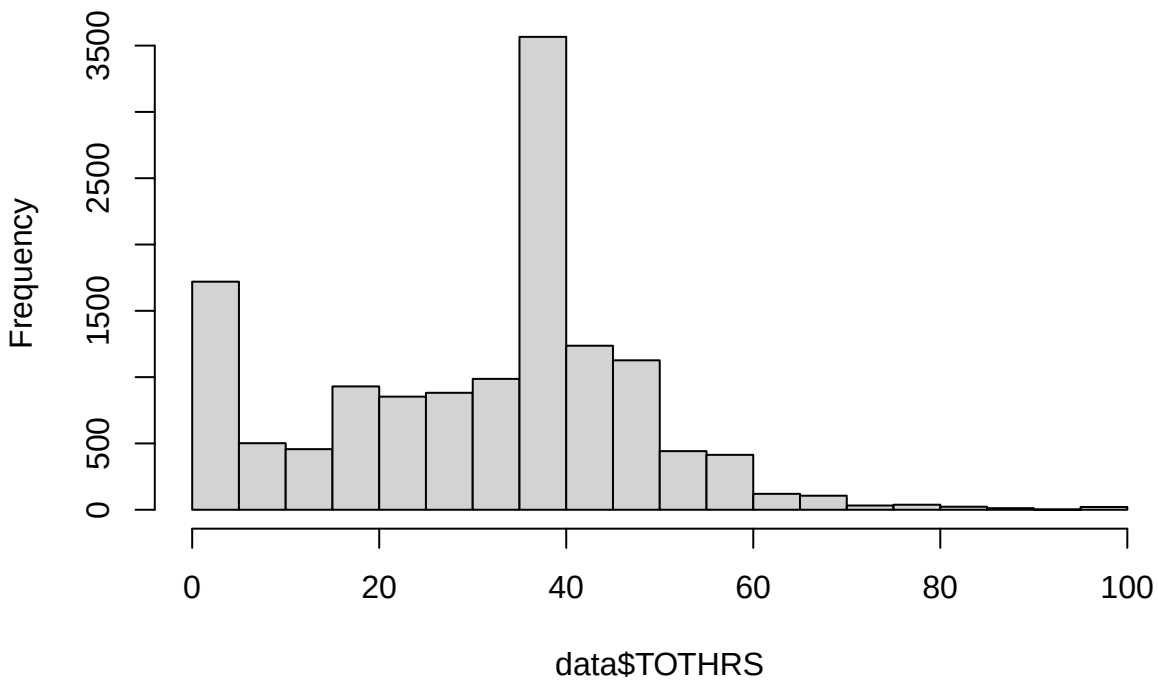


Let's draw a histogram for Working Hours

```
summary(data$TOTHRS)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.00   20.00   37.00   31.68   42.00   97.00    8957
```

```
hist(data$TOTHRS, main="Work hours per week")
```



```
##You can also embed plots, for example:
```

the Chi-squared test

```
chisq <- chisq.test(data$SEX, data$AGEEULR)
chisq
```

```
##
##  Pearson's Chi-squared test
##
## data:  data$SEX and data$AGEEULR
## X-squared = 23.159, df = 11, p-value = 0.01679
```

the test statistic is 23.159, with 11 degrees of freedom p-value is 0.01679 - probability of getting a more extreme test stat is 1.7% lower than 5% $\rightarrow$ reject H_0, the two variables are related

4. TESTING THE DIFFERENCE BETWEEN MEANS

H_0: men and women work the same number of hours on average
H_1: men and women work different amount of hours on average

the hours worked by men:

```
   hours_men <- data$TOTHRS[data$SEX==1]
```

the hours worked by women:

```
   hours_women <- data$TOTHRS[data$SEX==2]
```

```
meanstest <- t.test(hours_men, hours_women)
meanstest
```

```
##
##  Welch Two Sample t-test
##
## data:  hours_men and hours_women
## t = 33.062, df = 13412, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##   9.070153 10.213410
## sample estimates:
## mean of x mean of y
##  36.47820  26.83642
```

Mean for men is 36.48 and for women is 26.84.
P-value very close to 0 ($p - value < 2.2e^{-16}$) $\rightarrow$ reject H_0 - men and women work different number of hours

Note: we can use a t-test even when the sample size is large it gets very similar to a normal distribution

5. TESTING CORRELATIONS

H_0: hours worked and age are uncorrelated
H_1: hours worked and age are correlated

```
cortest <- cor.test(data$AGEEULR, data$TOTHRS)
cortest
```

```
##
##  Pearson's product-moment correlation
```

```
##
## data:  data$AGEEULR and data$TOTHRS
## t = -1.7702, df = 13469, p-value = 0.07671
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.03213017  0.00163619
## sample estimates:
##         cor
## -0.01525134
```

correlation is -0.015 (close to 0) p-value is $0.08 > 0.05 \rightarrow$ cannot reject H_0, we cannot rule out that the variables are uncorrelated.

Note: you should use two truly continuous variables for this test; age was ordinal in this case, which is not ideal but it was acceptable for a quick demonstration (the variable had equal intervals of 5 years, more than 10 categories, and actual age is really continuous.)

Example1: Scots Miners data

Download the data and read it into R

```
library(haven)
minerdata <- read.table("miners_data.csv", header = TRUE, sep=",", stringsAsFactors = FALSE)
```

Select two nominal/ordinal and two continuous variables from the dataset

Nominal:

```
marital <- minerdata$Marstat
parish <- minerdata$Par
```

Continuous:

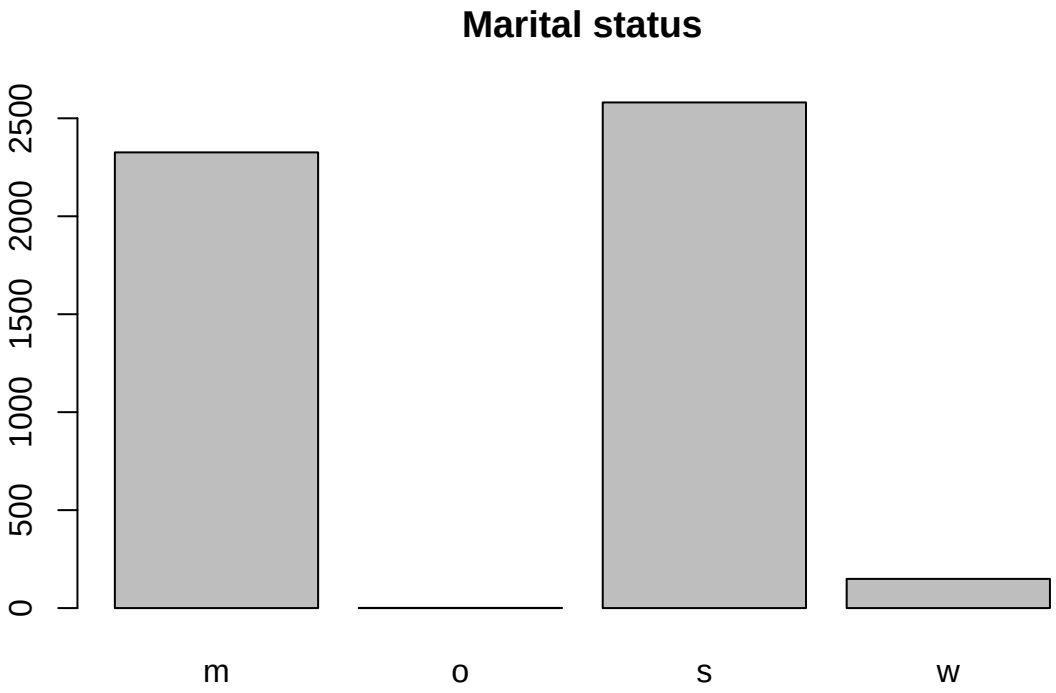
```
age <- minerdata$Age
count <- minerdata$Count # This variable may not be meaningful as it is the identifier of each case,
                         # we only use it for demonstration purpose.
```

Explore the variables

```
table(marital, useNA = "always")
```

```
## marital
##    m    o    s    w <NA>
## 2326    1 2581  149    0
```

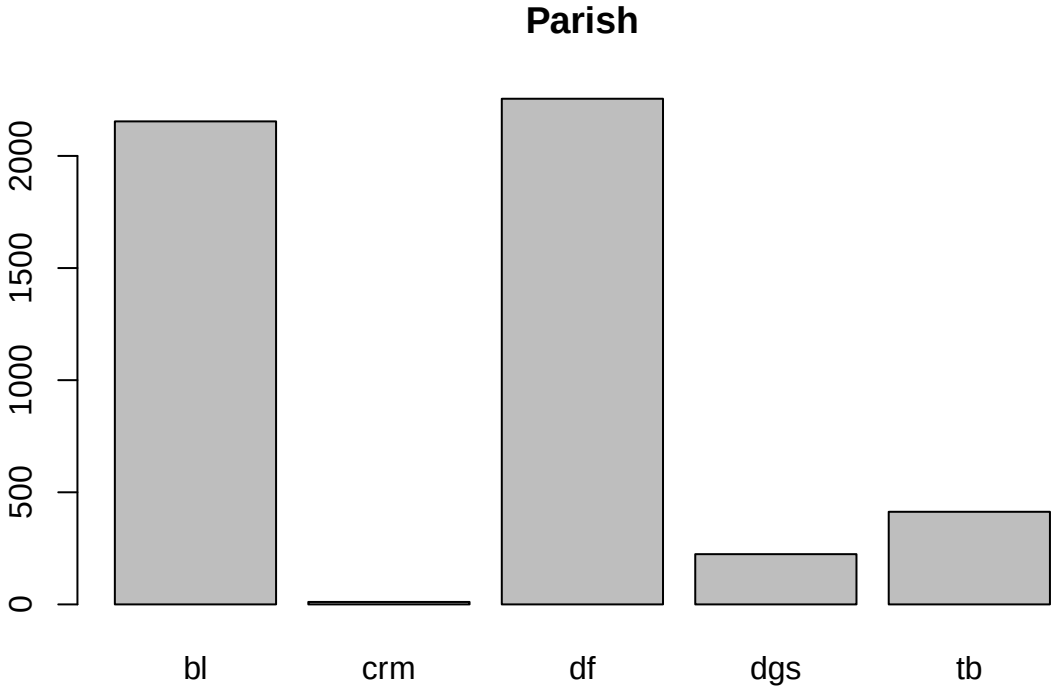
```
barplot(table(marital), main = "Marital status")
```



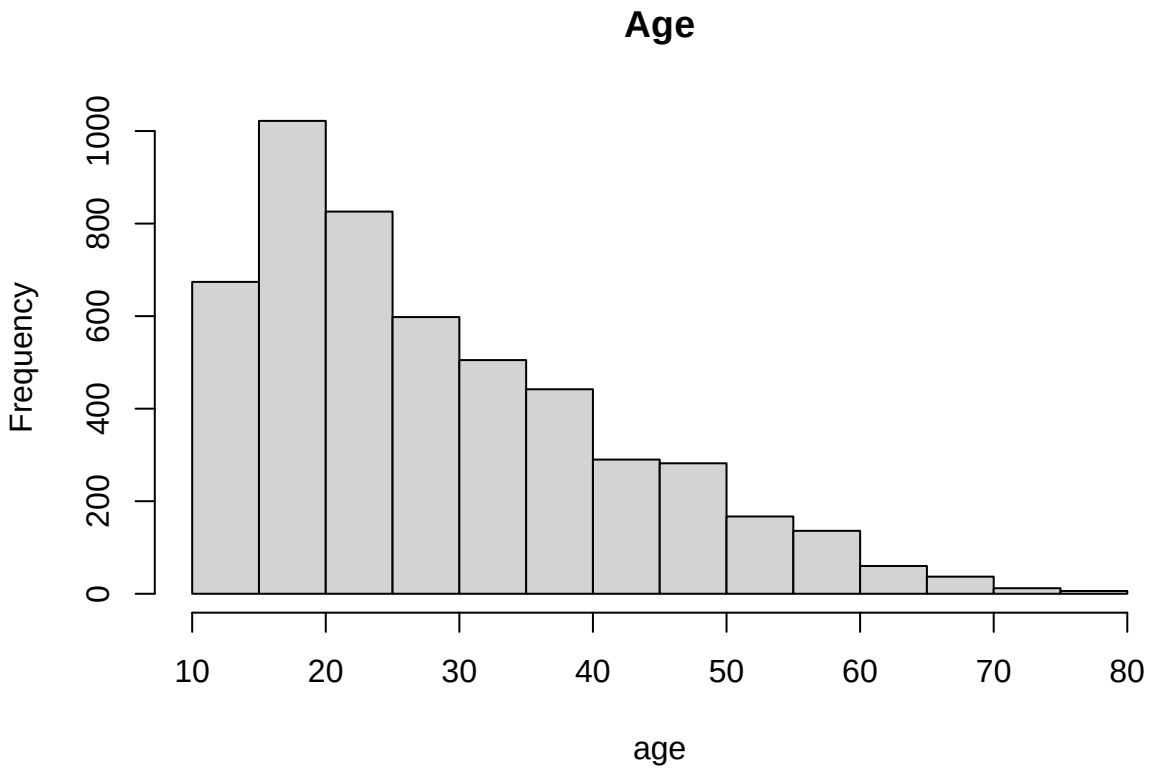

```
table(parish, useNA = "always")
```

```
## parish
##   bl  crm   df  dgs   tb <NA>
## 2154   11 2255  224  413    0
```

```
barplot(table(parish), main = "Parish")
```



```r
hist(age, main = "Age")
```



Select two nominal/ordinal variables and test their independence - chisq.test()

H_0: parish and marital are independent
H_1: pairsh ang marital are not independent

Contingency table:

```r
table(parish,marital)
```

```
##       marital
## parish    m    o    s    w
##    bl  1027    1 1063   63
##    crm    3    0    8    0
##    df  1038    0 1143   74
##    dgs   87    0  132    5
##    tb   171    0  235    7
```

Chi-squared test

```r
parmar <- chisq.test(parish,marital)
```

```
## Warning in chisq.test(parish, marital): Chi-squared approximation may be
## incorrect
```

```r
parmar
```

```
##
##  Pearson's Chi-squared test
##
## data:  parish and marital
```

```
## X-squared = 19.455, df = 12, p-value = 0.07811
```

The expected count for one cell is less than 1. Chi-sq test might not be suitable

Recode data

if you want to exclude the parish with fewer respondents, we can drop it. (you don't have to do this, it's up to your research questions. We only do this to demonstrate the data recoding)

```
parish4 <- parish
parish4[parish4=="crm"] <- NA
barplot(table(parish4), main = "Parish")
```




```
marital3 <- marital
marital3[marital3=="o"] <- NA
```

```
table(parish4, marital3)
```

```
##        marital3
## parish4    m    s    w
##     bl  1027 1063   63
##     df  1038 1143   74
##     dgs   87  132    5
##     tb   171  235    7
```

```
parmar <- chisq.test(parish4,marital3)
parmar
```

```
##
##  Pearson's Chi-squared test
##
## data:  parish4 and marital3
## X-squared = 15.918, df = 6, p-value = 0.0142
```

the test statistic is 15.918, with (4-1)*(3-1)=6 degrees of freedom.

P-value is 0.0142 - probability of getting a more extreme test stat is 1.4% lower than 5% → reject H_0, the two variables are related (more single people were in dgs and tb)

TESTING THE DIFFERENCE BETWEEN MEANS - t.test()

H_0: single and married people have the same age
H_1: single and married people have the different age

Age of single people:

```
age_s <- age[marital=="s"]
```

Age of married people

```
age_m <- age[marital=="m"]
```

```
agemar <- t.test(age_s,age_m)
agemar
```

```
##
##  Welch Two Sample t-test
##
## data:  age_s and age_m
## t = -63.084, df = 4001.1, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -17.99931 -16.91425
## sample estimates:
## mean of x mean of y
##  20.23712  37.69390
```

Mean of age for single is 20.2, and for married is 37.7.
P-value is $2.2e^{-16}$, lower than 5% -→ reject H_0 - Age of single and married people are different.

Select two continues variables and test if their correlation is zero - cor.test()

H_0: count and age are uncorrelated
H_1: count and age are correlated

```
age_count <- cor.test(age, count)
age_count
```

```
##
##  Pearson's product-moment correlation
##
## data:  age and count
## t = 1.3762, df = 5055, p-value = 0.1688
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.008213815  0.046890789
## sample estimates:
##        cor
## 0.01935318
```

correlation is 0.019 (close to 0) p-value is $0.17 > 0.05$ → cannot reject H_0, we cannot rule out that the variables are uncorrelated

Example2: Scottish Social Attitudes Survey, 2016

1. Look for a dataset

2. Download the data and read it into R

```
ssadata <- read.table("ssa2016_for_ukda.tab", header = TRUE)
dim(ssadata)
```

```
## [1] 1237  386
```

```
names(ssadata)
```

```
##   [1] "pserial"   "OddEven"   "rsex"      "rage"      "Househld"  "RAgeCat"
##   [7] "RAgeCat2"  "RSexAge"   "RSexAge2"  "MarStat6"  "MarStat"   "Married"
##  [13] "NumCh"     "NCh415"    "NCh318"    "HhCh04"    "HhCh511"   "HhCh1215"
##  [19] "HhCh1617"  "RCh04"     "RCh511"    "RCh1215"   "RCh1617"   "NumAd"
##  [25] "HHType"    "rage1824"  "rage1829"  "ragecat3"  "ragecat4"  "rage1624"
##  [31] "rage1629"  "RSexage3"  "RSexage4"  "children"  "childsch"  "hhtypeB"
##  [37] "REconAct"  "RInEduc"   "RInGvTrn"  "RInPaidW"  "RWaitPdW"  "RUnemReg"
##  [43] "RUnemLok"  "RUnemNlk"  "RSickDis"  "RRetired"  "RLookAfH"  "REAElse"
##  [49] "REconSum"  "R2OpCat"   "R2Class"   "R2ClasGp"  "R2EconPs"  "R2JbHrCI"
##  [55] "R2PartFI"  "REmplyee"  "SEconAct"  "SInEduc"   "SInGvTrn"  "SInPaidW"
##  [61] "SWaitPdW"  "SUnemReg"  "SUnemLok"  "SUnemNlk"  "SSickDis"  "SRetired"
##  [67] "SLookAfH"  "SEAElse"   "SEconSum"  "P2OpCat"   "P2Class"   "P2ClasGp"
##  [73] "P2EconPs"  "P2JbHrCI"  "P2PartFI"  "SReadpap"  "WhPaper"   "WebNews"
##  [79] "WNwSite1"  "WNwSite2"  "paper2"    "SupParty"  "ClosePty"  "PartyFW"
##  [85] "PartyIdS"  "PtyAllgS"  "Idstrng"   "Politics"  "party3"    "idstrng2"
##  [91] "politic2"  "ScotPar2"  "ScotPar3"  "ECPolic2"  "NatID"     "NHSSat"
##  [97] "SNHSIE12"  "SNHSWh12"  "SNHSIE13"  "SLivIE12"  "SLivWh12"  "slivie13"
## [103] "EconIE12"  "EconWh12"  "econie13"  "ExecDo2"   "ExecList"  "UKList"
## [109] "LCList"    "UKIntNat"  "NatInNat"  "UKFairD"   "SEFairD"   "CounFair"
## [115] "DoesInfl"  "OughInfl"  "VoiceUK3"  "VoiceUK4"  "VoiceUKC"  "SayInGv3"
## [121] "SayInGv4"  "SayInGvC"  "DoNoneS"   "DoMPS"     "DoGovS"    "DoLGS"
## [127] "DoRespS"   "DoMtgs"    "DoTVS"     "DoSignS"   "DoRaisS"   "DoProtS"
## [133] "DoEventS"  "DoSpkS"    "DoGrpS"    "DoJoinS"   "DoParty"   "DoActivS"
## [139] "DoMoneyS"  "Indytalk"  "Electalk"  "EUtalk"    "Indyres"   "indyresa"
## [145] "IndyresB"  "IndyresC"  "MaleObes"  "FemObes"   "WorrOWt"   "WtInher"
## [151] "WtMetab"   "WtEat"     "WtTypFoo"  "WtExer"    "ObHlthD1"  "ObHlthD2"
## [157] "ObHlthD3"  "ObHlthD4"  "ObHlthD5"  "ObHlthD6"  "ObHlthD7"  "ObHlthD8"
## [163] "ObHlthD9"  "ObHlthD10" "ObHlthD11" "ObHlthD12" "ObHlthD14" "ObHlthE1"
## [169] "ObHlthE2"  "ObHlthE3"  "ObHlthE4"  "ObHlthE5"  "ObHlthE6"  "ObHlthE7"
## [175] "ObHlthE8"  "ObHlthE9"  "ObHlthE10" "ObHlthE11" "ObHlthE12" "ObHlthE14"
## [181] "ObHlthF1"  "ObHlthF2"  "ObHlthF3"  "ObHlthF4"  "ObHlthF5"  "ObHlthF6"
## [187] "ObHlthF7"  "ObHlthF8"  "ObHlthF9"  "ObHlthF10" "ObHlthF11" "ObHlthF12"
## [193] "ObHlthF14" "HFExpen"   "HFTime"    "PhysTim2"  "PhysExp"   "FFCheap"
## [199] "DietVEx"   "HarmObes"  "HarmSmok"  "HarmAlc"   "RespObD1"  "RespObD2"
## [205] "RespObD3"  "RespObD4"  "RespObD5"  "RespObD6"  "RespObD7"  "RespObD8"
## [211] "RespObD9"  "RespObD10" "RespObD11" "RespObD13" "RespObE1"  "RespObE2"
## [217] "RespObE3"  "RespObE4"  "RespObE5"  "RespObE6"  "RespObE7"  "RespObE8"
## [223] "RespObE9"  "RespObE10" "RespObE11" "RespObE13" "RespObF1"  "RespObF2"
## [229] "RespObF3"  "RespObF4"  "RespObF5"  "RespObF6"  "RespObF7"  "RespObF8"
## [235] "RespObF9"  "RespObF10" "RespObF11" "RespObF13" "FatTax2"   "SugTaxFo"
## [241] "SugTaxDk"  "FlavDrin"  "RedSize2"  "PromChld"  "SportSpo"  "Reform"
## [247] "PricePro"  "BanFtAd2"  "BanSugAd"  "BanPosit"  "FFoutlet"  "MoreWMC"
## [253] "FFSPos"    "OwnShape"  "BetterHl"  "ProbHlth"  "ProbHlt2"  "ProbHlt3"
```

```
## [259] "ProbHlthC" "MoneyHlt"  "MoneyHl2"  "MoneyHl3"  "MoneyHlC"  "ChoiCirs"
## [265] "RichWork"  "RichJob"   "RichInju"  "IncUnf"    "IncomGap"  "BecGenes"
## [271] "BecHouse"  "BecLearn"  "BecWork"   "BecPoor"   "BecInjus"  "AnyChoic"
## [277] "HlthGap"   "RespoHlt"  "EURefV"    "ActEURfV"  "Voted2"    "VoteC3"
## [283] "VoteL3"    "ScWBrit"   "Vote1st3"  "Vote2nd3"  "Refvot16"  "SCAccept"
## [289] "TaxWhole"  "TaxPoor"   "ImVotSP"   "ImVotUK"   "ImVotLoc"  "ReDistrb"
## [295] "BigBusnn"  "Wealth"    "RichLaw"   "Indust4"   "TradVals"  "StifSent"
## [301] "DeathApp"  "Obey"      "WrongLaw"  "Censor"    "ViewWt"    "HappyWt"
## [307] "TenureNW"  "Tenure2"   "tenure3"   "HIncDif4"  "SRHlthSG"  "DisNew3"
## [313] "HowTall"   "HowTallF"  "HowTallI"  "HowTallC"  "EstHt"     "MuchWgh"
## [319] "MuchWghP"  "MuchWghS"  "MuchWgh2"  "MuchWghK"  "EstWt"     "RelgSumS"
## [325] "FrlSumS"   "ChAttend"  "NatBrit"   "NatEng"    "NatEuro"   "NatIrish"
## [331] "NatNI"     "NatScot"   "NatUlst"   "NatWelsh"  "NatAsia"   "NatAfric"
## [337] "NatOth"    "NatNone"   "BNationU"  "BestNatU"  "bestnat2"  "RaceOri3"
## [343] "SchQual"   "PschQual"  "HEdQual"   "HEdQual2"  "hedband"   "VoteUKGE"
## [349] "Vote15"    "Vote11c"   "Vote11L"   "DecScotA"  "JbType"    "r2lastjb"
## [355] "r2emplye"  "r2superv"  "r2many"    "r2supmn2"  "R2OcSec2"  "R2EmpWrk"
## [361] "R2SICgp"   "R2OCSEC3"  "P2LastJb"  "P2emplye"  "P2superv"  "P2SICGp"
## [367] "P2Many"    "P2OcSec2"  "P2EmpWrk"  "p2OCSEC3"  "MainInc4"  "HHIncome"
## [373] "INCQURT2"  "incqurt3"  "SRHlthLR"  "incsour"   "leftrigh"  "libauth"
## [379] "libauth2"  "leftrig2"  "urbanac"   "point"     "stratum"   "wtfactor"
## [385] "oldwt"     "qsimd16"
```

3. Select two nominal/ordinal and two continuous variables from the dataset

Nominal:

```
party <- ssadata$PartyIdS  #Party identification (compressed)
scotsind <- ssadata$Refvot16 #How would you vote in a Scottish Independence referendum if held?
```

Continuous:

```
age <- ssadata$rage
rweight <- ssadata$EstWt  # How much do you weigh? (converted to kg)
```

Let's explore the variables

```
table(party, useNA = "always")
```

```
## party
##    1    2    3    4    5    6    7    8    9 <NA>
##  224  227   50  439   25  143   23    3  103    0
```

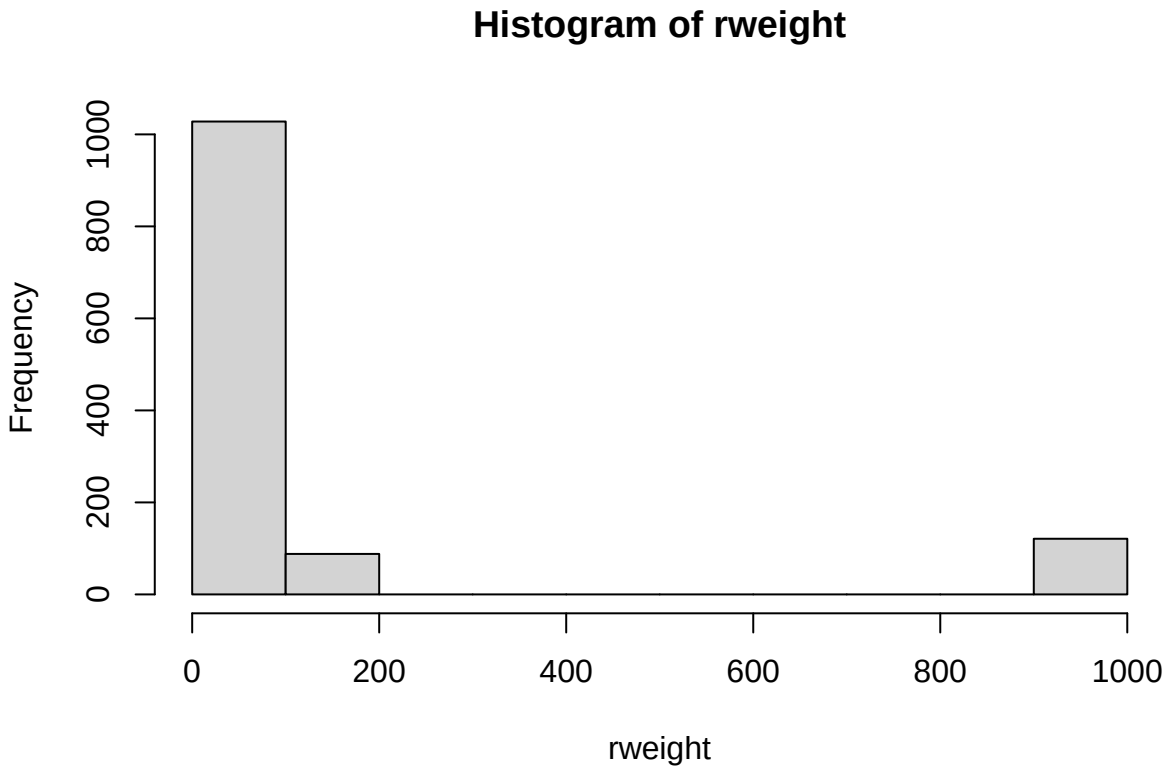
```
table(scotsind, useNA = "always")
```

```
## scotsind
##    1    2    3    4    8    9 <NA>
##  491  604   94   38    7    3    0
```

```
hist(age)
```



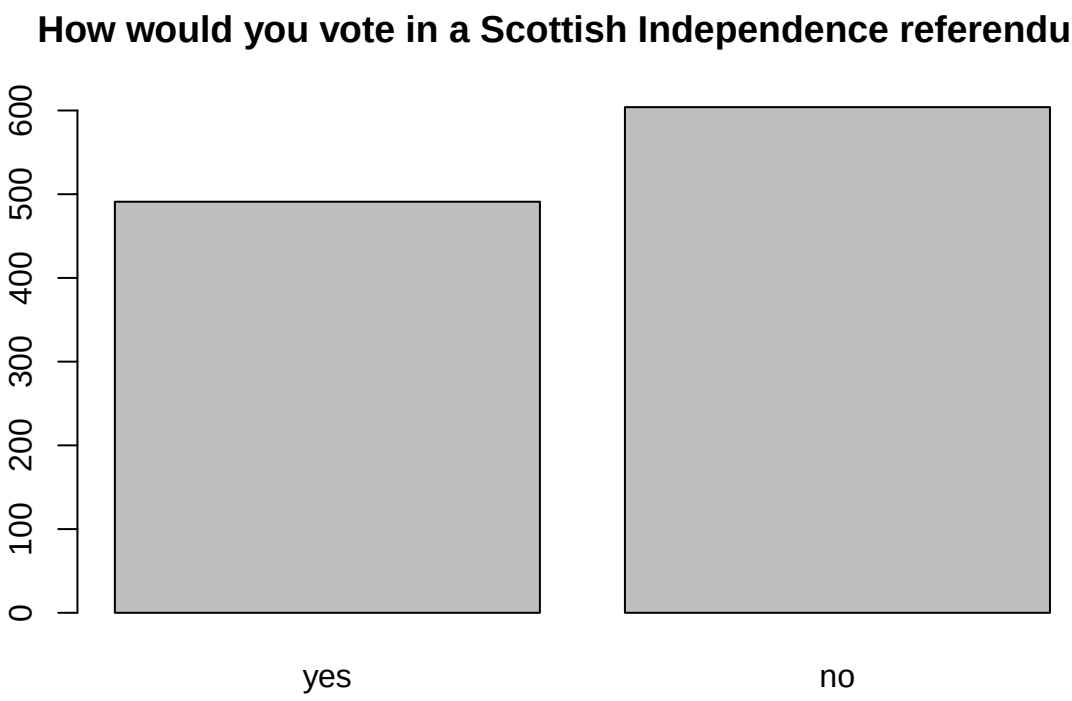

```
hist(rweight)
```



Recoding and addressing missingness

For scotsind, only keep 1-"yes" 2-"no". Other values indicate different types of missingness

```
scotsind2 <- scotsind
scotsind2[scotsind %in% c(3,4,8,9)] <- NA
barplot(table(scotsind2), main = "How would you vote in a Scottish Independence referendum?",
        names.arg = c("yes","no"))
```

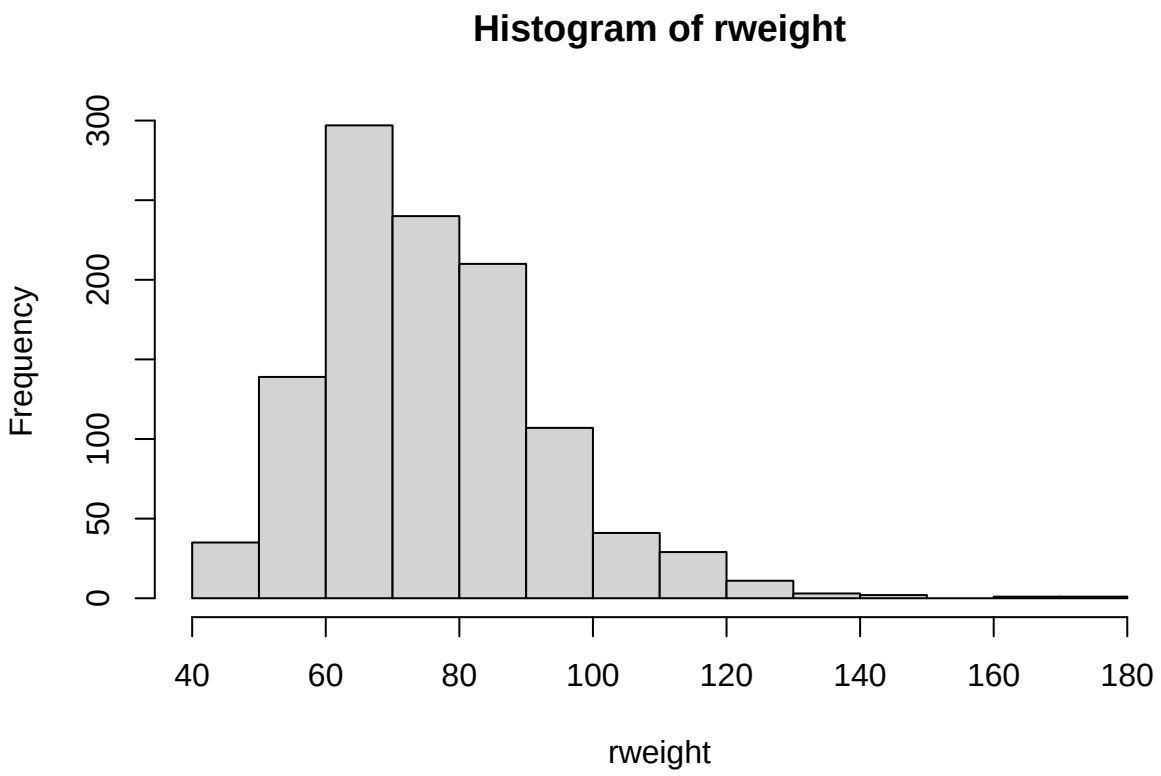


For rweight, missing values are set as 998 and 999

```
rweight[rweight>=998]<-NA
summary(rweight) #121 cases convert to NA
```

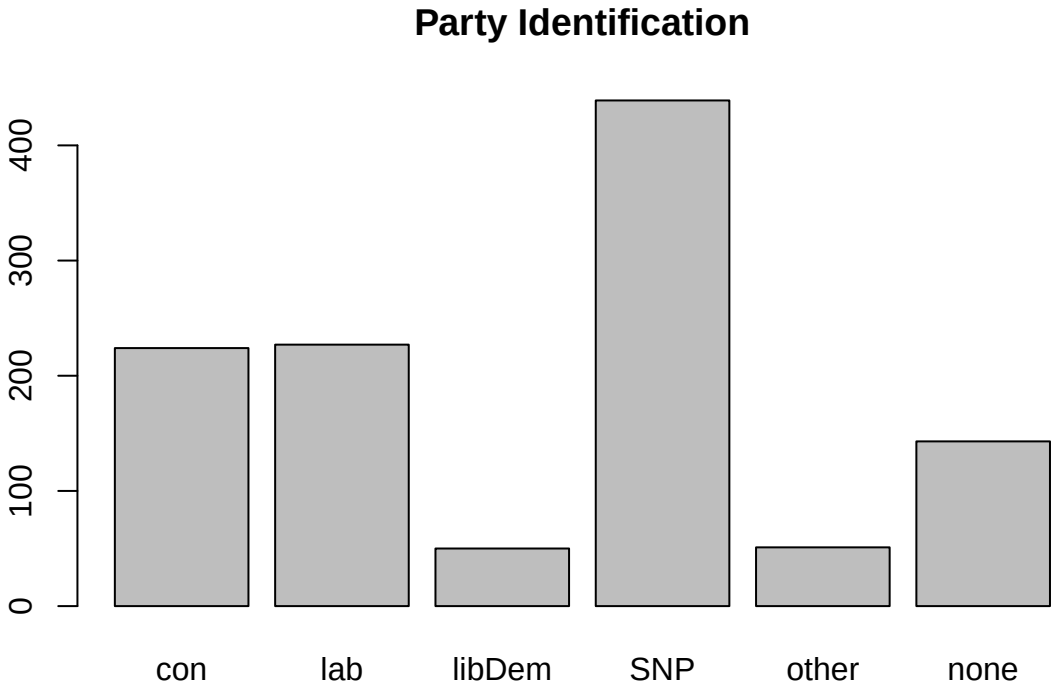
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   40.00   63.60   74.50   76.29   85.80  179.00     121
```

```
hist(rweight)
```



For party, some groups have few people, collapse them into a larger group → other party

```
party[party==9] <- NA
party[party %in% c(5,7,8)] <- 5
barplot(table(party), main = "Party Identification",
        names.arg = c("con","lab","libDem","SNP","other", "none"))
```



4. Select two nominal/ordinal variables and test their independence - chisq.test()

H_0: party identity and the voting for scottish independence are independent
H_1: party identity and the voting for scottish independence are not independent

```
round(prop.table(table(scotsind2,party),2),2)
```

```
##          party
## scotsind2    1    2    3    4    5    6
##         1 0.07 0.26 0.10 0.80 0.51 0.43
##         2 0.93 0.74 0.90 0.20 0.49 0.57
```

```
scotsparty <- chisq.test(scotsind2,party)
```

fisher.test(scotsind, party, simulate.p.value=TRUE) #Fisher's exact test is a statistical significance test used in the analysis of contingency tables, particularly when sample sizes are small.

```
scotsparty
```

```
##
##  Pearson's Chi-squared test
##
## data:  scotsind2 and party
## X-squared = 377.09, df = 5, p-value < 2.2e-16
```

the test statistic is 377.09, with 5 degrees of freedom p-value is very close to 0, lower than 5% reject H_0: the two variables are related

5. Testing the difference between means- t.test()

H_0: people who vote yes and who vote no for Scottish independence have same age on average
H_1: people who vote yes and who vote no for Scottish independence have different age on average

age of those who vote yes for Scottish independence

```
age_yes <- age[scotsind2==1]
```

age of those who vote no for Scottish independence

```
age_no <- age[scotsind2==2]
```

```
meanstestage <- t.test(age_yes, age_no)
meanstestage
```

```
##
##  Welch Two Sample t-test
##
## data:  age_yes and age_no
## t = -9.6227, df = 1055, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -12.124109  -8.017014
## sample estimates:
## mean of x mean of y
##  47.01222  57.08278
```

mean age for voting for yes: 47.0, mean for voting for no: 57.1 p-value is very close to 0, lower than 5% $\rightarrow$ reject H_0: people who vote yes and who vote no for Scottish independence have different age on average.

6. Select two continuous variables and test if their correlation is zero - cor.test()

H_0: weight and age are uncorrelated
H_1: weight and age are correlated

```
corweightage <- cor.test(rweight, age)
corweightage
```

```
##
##  Pearson's product-moment correlation
##
## data:  rweight and age
## t = 0.25949, df = 1114, p-value = 0.7953
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.05093048  0.06642547
## sample estimates:
##         cor
## 0.007774267
```

correlation is 0.008 (close to 0) p-value is 0.7953$>$ 0.05 $\rightarrow$ cannot reject H_0, we cannot rule out that the variables are uncorrelated.